

PORAMAT SANGSAENGSAI

095-717-9316 | Poramat.Sang@gmail.com | linkedin.com/in/poramat-sang

SUMMARY

Electrical & Electronics Engineer with end-to-end expertise in Circuit Design, PCB Layout, Embedded Firmware, IoT Cloud Integration, and AI-Driven Web Development, with 7+ hardware and software projects built independently, ready to deliver real-world impact as an engineer.

SKILLS

Technical Skills:

- **Hardware & Power Systems:** Electrical & Electronic Engineering, PCB Engineering & Development, Microcontroller Design & Development, Energy System Integration, Measuring Instruments
- **Embedded, IoT & AI Systems:** Embedded Systems, Internet of Things (IoT), Data Acquisition & Logging, System Testing, Artificial Intelligence (AI)

Programming: C, C++, Python, HTML, CSS, JavaScript

Software: Microsoft Office, EasyEDA, Arduino IDE, ESP-IDF, LabVIEW, MATLAB

EDUCATION

Kasetsart University

Jan 2022 – Dec 2024

Master of Engineering – Electrical Engineering | GPA: 3.83 / 4.00

Thammasat University

Aug 2017 – May 2021

Bachelor of Science – Electronics Physics | GPA: 2.62 / 4.00

RESEARCH & PROJECT EXPERIENCE

Raspberry Pi-Controlled Low-Cost Lock-In Amplifier (2020–2021)

- Designed and developed an end-to-end lock-in amplifier system by integrating fundamental analog circuitry with Raspberry Pi control.
- Achieved high-precision signal detection for optical applications, delivering performance comparable to commercial-grade instruments.
- Optimized for a compact form factor and significantly lower cost without compromising measurement accuracy.

Technologies: Raspberry Pi, Python, Analog Circuit Design, Signal Processing, Lock-in Detection, Optical Sensing

IoT-Based Single-Phase Power Measurement for Hydro Turbines (2022–2024)

- Developed an end-to-end monitoring solution integrating microcontroller-based hardware with multi-platform data visualization (PC & Smartphone).
- Implemented real-time data streaming via Blynk IoT application, enabling seamless remote monitoring and performance tracking.
- Achieved measurement accuracy comparable to commercial-grade power meters while optimizing system cost and scalability.

Technologies: ESP8266, C++, Arduino IDE, Blynk IoT, ADC, Current/Voltage Sensing, Data Logging

Three-Phase Power Measurement & Data Logging System (2024)

- Designed and developed an end-to-end three-phase power measurement system controlled by a high-performance microcontroller, supporting both single-phase and three-phase monitoring.
- Implemented comprehensive data logging and visualization on PC and smartphones, enabling real-time analysis of power parameters.
- Delivered measurement precision and calculation accuracy comparable to commercial-grade power analyzers, providing reliable data for industrial-level energy management.

Technologies: ESP8266, ESP-IDF, C++, Blynk IoT, Three-Phase Power Analysis, CT Sensor, Data Logging, UART

Custom ESP32 Microcontroller Development Board (2025)

- Designed and developed a custom, high-reliability ESP32-based microcontroller board from schematic capture to PCB layout, optimizing for ultra-low power consumption.
- Implemented comprehensive circuit protection mechanisms across all subsystems, ensuring robust hardware defense against overcurrent, overvoltage, and reverse polarity.
- Selected and sourced industrial-grade components to maximize energy efficiency and power management for long-term deployment and sustainability.

Technologies: *ESP32, EasyEDA, PCB Layout, Schematic Capture, Circuit Protection, Low-Power Design, SMD Components, Microcontroller*

Tuya Smart Home API Integration & Custom Control System (2025)

- Engineered a custom integration framework for Tuya Smart Home ecosystems by implementing secure OAuth API token authentication.
- Enabled advanced data point telemetry monitoring and executed complex device control actions outside the constraints of standard application features.
- Optimized communication latency and security between local software interfaces and Cloud APIs for stable smart home automation.

Technologies: *Tuya API, Python, OAuth 2.0, REST API, IoT Protocol, Cloud API, Local Network Integration*

Secure Private Cloud Storage System with AI File Management (2026)

- Developed an end-to-end private cloud storage solution featuring a custom-designed web interface with secure OAuth 2.0 multi-account authentication.
- Integrated external high-speed SSD storage for reliable data logging, high-capacity file hosting, and efficient read/write operations.
- Deployed local AI algorithms to enhance system security protocols and automate file organization based on file extensions and metadata.

Technologies: *Python, HTML, CSS, JavaScript, OAuth 2.0, Web Development, SSD Storage, AI-based Security, AI File Classification, Local Server, Metadata Processing*

MoA-Based AI Assistant Chatbot (2026)

- Architected and deployed a conversational AI chatbot leveraging the Mixture of Agents (MoA) methodology to synthesize responses from multiple LLMs.
- Integrated Gemini and Groq APIs to build a high-performance backend, orchestrating agent collaboration to achieve superior accuracy and contextual understanding.
- Designed an intuitive text-based chat interface with seamless state management for real-time, interactive user experiences.

Technologies: *Python, HTML, CSS, JavaScript, Mixture of Agents (MoA), Gemini API, Groq API, Llama, LLM Orchestration, State Management*